

BAYAN MOD, China

WMO: 524950

Lat: 40.750N

Lon: 104.500E

Elev: 4360

StdP: 12.52

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-5.0	-26.1	1.5	7.0	-21.9	1.9	8.2	25.2	26.4	22.2	22.9	2.1	40	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	93.1	60.8	90.6	59.8	88.2	58.9	66.6	81.8	64.7	80.0	63.1	78.2	10.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.1	98.3	71.2	60.0	91.0	70.1	57.9	84.2	70.0	33.9	82.2	32.2	79.9	31.0	78.0	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.5	22.7	19.7	DB	-16.2	98.2	5.4	3.3	-20.1	100.6	-23.2	102.5	-26.3	104.3	-30.2	106.7	(4)
(5)			WB	-16.6	68.4	5.4	2.2	-20.4	70.0	-23.6	71.2	-26.6	72.4	-30.5	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	13.3	21.7	35.3	50.6	62.1	72.0	76.3	72.8	62.1	47.3	30.9	17.0		
	DBStd	23.10	7.99	9.03	9.66	8.83	7.37	5.66	5.16	5.52	6.74	8.02	8.54	8.13		
	HDD50	4238	1137	792	461	97	6	0	0	0	3	146	574	1022		
	HDD65	7521	1602	1212	920	436	142	10	0	7	131	550	1024	1487		
	CDD50	3111	0	0	6	115	381	660	816	706	366	61	0	0		
	CDD65	919	0	0	0	5	52	220	351	247	43	1	0	0		
	CDH74	10021	0	0	0	95	723	2458	3860	2433	437	15	0	0		
	CDH80	3942	0	0	0	15	187	957	1768	921	94	0	0	0		
	WSAvg	7.9	6.1	7.1	8.7	9.7	9.9	8.9	8.3	7.7	7.1	6.9	7.3	6.6		
	PrecAvg	3.5	0.0	0.0	0.1	0.1	0.3	0.5	1.0	1.0	0.4	0.2	0.0	0.0		
(15) Precipitation	PrecMax	7.8	0.2	0.1	0.4	0.6	1.2	1.7	3.4	2.8	1.2	0.8	0.7	0.2		
	PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.5	0.0	0.0	0.1	0.2	0.3	0.5	0.7	0.7	0.3	0.2	0.2	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	51.7	68.1	82.0	88.5	94.5	98.7	94.2	88.6	76.2	59.3	43.9		
		MCWB	29.6	35.1	44.3	51.0	54.7	58.1	64.7	62.8	58.0	50.7	40.3	30.8		
	2%	DB	36.1	45.9	61.6	76.4	84.8	91.1	94.3	90.9	82.2	70.6	54.5	38.9		
		MCWB	26.2	32.1	40.6	48.2	53.3	57.8	61.6	60.7	55.8	48.2	38.1	28.5		
	5%	DB	31.7	42.0	57.1	71.8	81.6	88.5	91.7	88.1	78.8	66.7	50.1	34.8		
		MCWB	23.5	30.0	38.4	46.2	52.0	56.9	61.0	59.8	54.7	46.7	35.8	26.1		
	10%	DB	27.8	38.0	52.5	67.6	78.0	85.8	89.1	85.4	75.6	62.6	45.9	31.0		
		MCWB	21.0	27.7	35.9	44.8	50.3	56.2	60.4	59.5	53.9	44.5	33.9	23.6		
	0.4%	WB	29.8	36.3	44.9	52.9	58.1	64.5	69.2	69.1	63.8	53.5	41.8	31.6		
		MCDB	41.4	50.0	65.9	73.5	79.3	79.6	90.0	80.5	75.8	69.1	55.3	43.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.6	32.6	41.4	50.2	55.4	62.3	66.9	66.8	60.7	50.4	39.0	28.8		
		MCDB	35.5	44.2	59.1	70.8	77.6	76.9	84.6	79.6	73.7	66.1	52.0	38.6		
	5%	WB	23.9	30.4	39.0	47.9	53.5	60.4	65.0	64.8	58.4	48.0	36.6	26.2		
		MCDB	31.4	41.2	54.5	68.5	76.3	77.2	81.2	79.4	71.1	63.2	49.5	34.4		
	10%	WB	21.2	28.0	36.7	45.5	51.8	59.0	63.6	62.9	56.3	45.6	34.1	23.7		
		MCDB	27.7	37.7	51.6	66.4	74.7	77.8	79.3	77.3	70.2	60.0	45.5	30.7		
	5% DB	MDBR	25.0	25.9	24.9	24.0	23.3	21.6	20.5	20.4	21.0	23.5	23.1	23.5		
		MCDBR	29.3	30.1	30.2	29.7	27.6	25.6	24.2	23.6	25.2	28.1	27.7	26.6		
		MCWBR	21.2	19.7	16.8	14.0	11.6	8.4	7.0	7.2	9.2	13.1	15.9	18.7		
		MCDBR	28.3	28.1	27.3	26.2	23.1	19.5	19.2	18.6	19.3	24.7	25.9	25.7		
	5% WB	MCWBR	20.7	18.5	15.4	12.5	10.5	7.0	6.3	6.6	8.2	12.5	15.2	18.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.353	0.401	0.449	0.472	0.481	0.471	0.452	0.422	0.373	0.328	0.302		
	taud		2.163	2.039	1.923	1.844	1.846	1.891	1.980	2.015	2.029	2.082	2.136	2.149		
	Ebn at Noon		265	265	265	260	258	255	256	257	256	254	251	253		
	Edh at Noon		32	43	54	63	64	62	56	53	48	41	33	30		
(44) All-Sky Solar Radiation	RadAvg		827	1179	1596	1979	2222	2200	2041	1909	1614	1320	930	756		
	RadStd		67	55	110	94	102	137	110	91	93	52	55	57		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page